Exercice 4 : let $a \in \mathbb{R}^n$, $n \in \mathbb{N}$

The solution of $\min_{w} \| Xw - y \|^2 + \lambda \| w \|^2$ is :

$$w = (X^T X + \lambda I)^{-1} X^T y$$

let $w = X^T a$ then $(X^T X + \lambda I) X^T a = X^T y$

$$\iff X^T (X X^T + \lambda I) a = X^T y$$

$$\iff (X^T X + \lambda I) a = y$$

the kernel matrix $K = X X^T$ so $a = (K + \lambda I)^{-1} y$

finally the kernelized ℓ_2-RLS is :

$$f(x) = w^T x = (X^T a) x = a^T X x$$

$$\boxed{f(x) = \sum_{i=1}^{n} a_i \, k(X_i, x)}$$